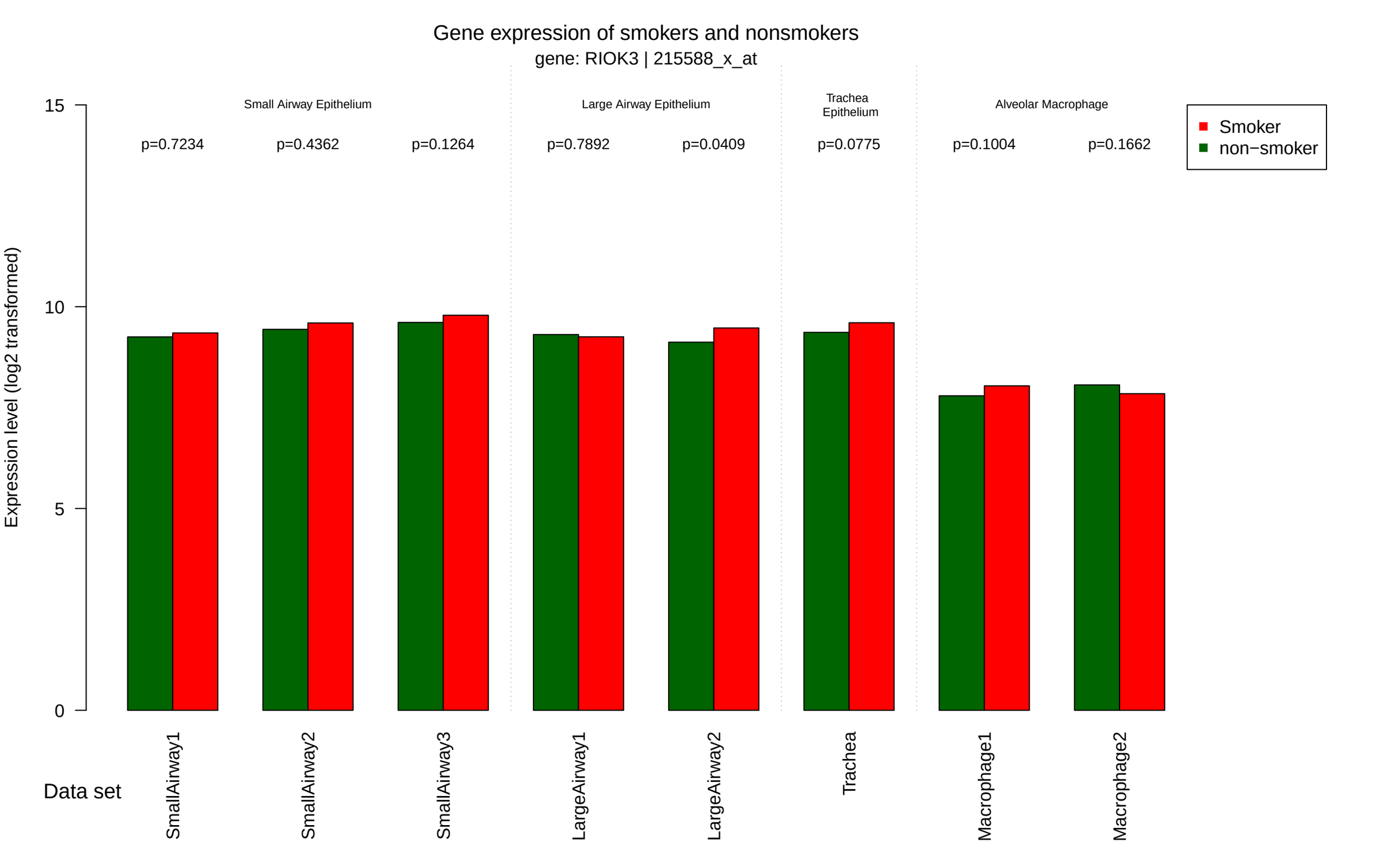
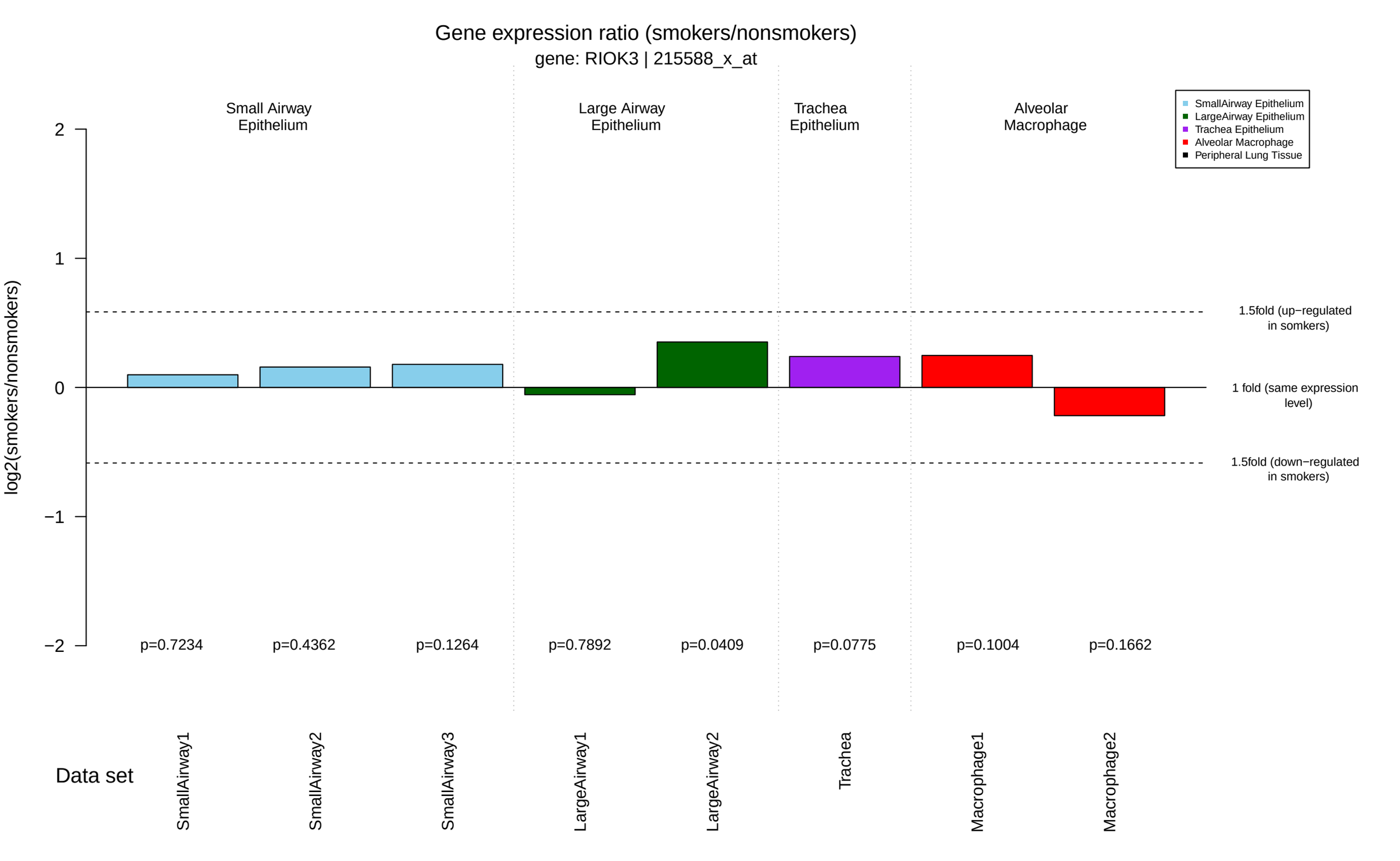
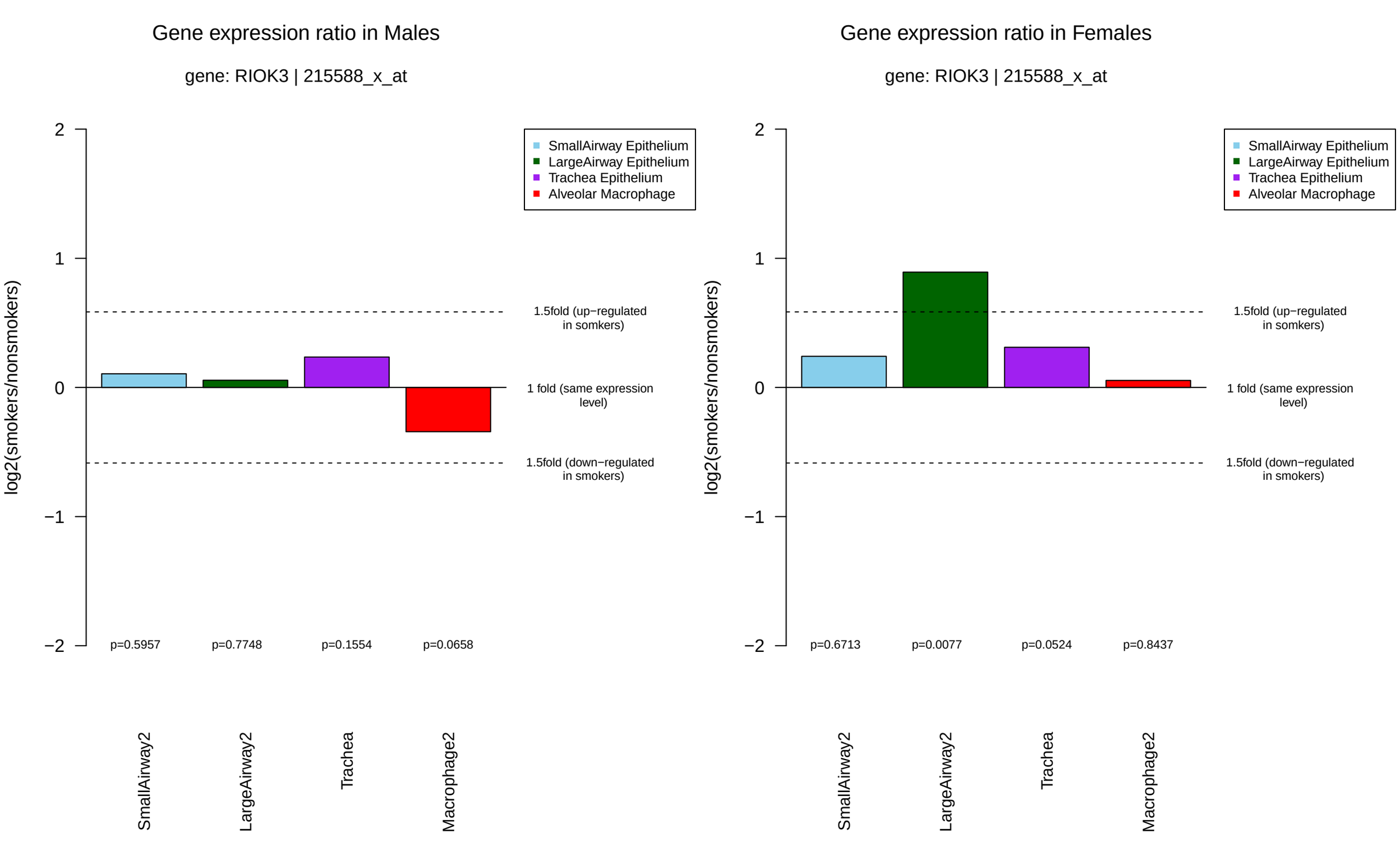
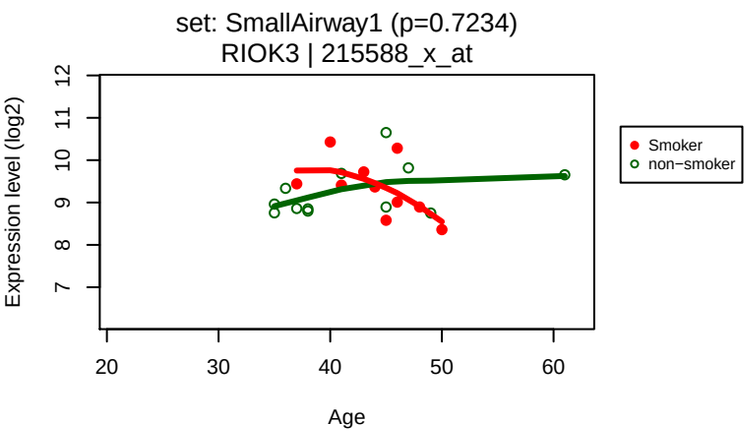
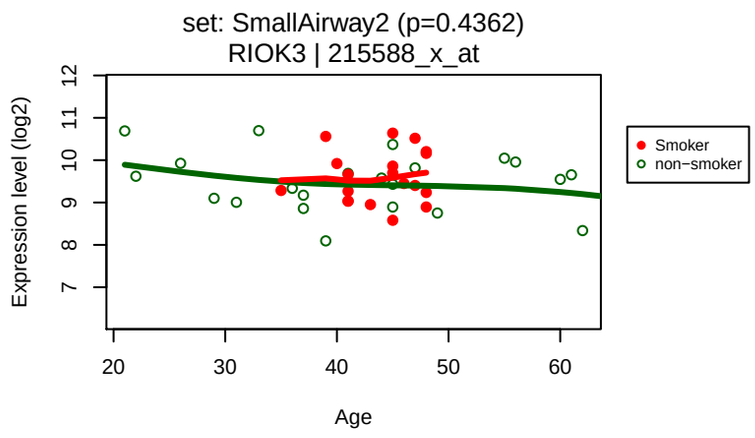
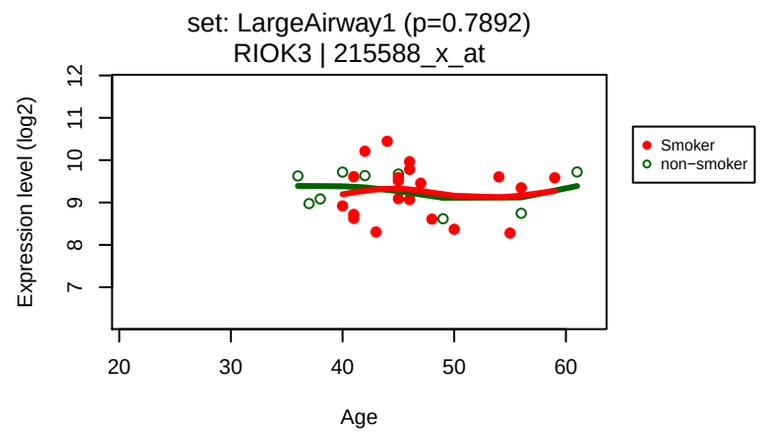

Gene expression of smokers and nonsmokers
gene: RIOK3 | 215588_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7234
p=0.4362
p=0.1264
p=0.7892
p=0.0409
p=0.0775
p=0.1004
p=0.1662
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

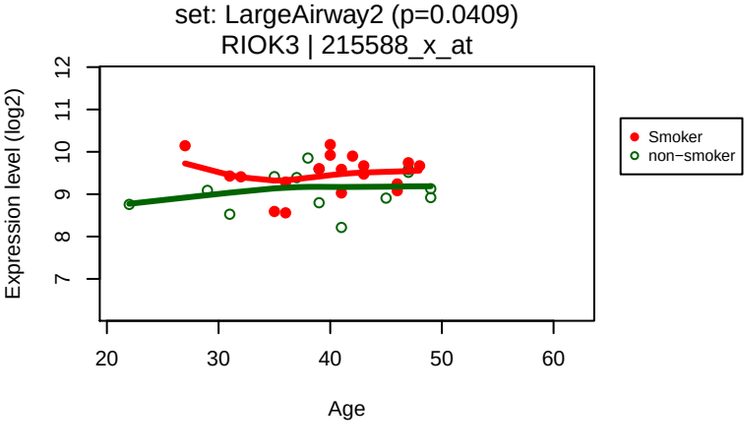

Gene expression ratio (smokers/nonsmokers)
gene: RIOK3 | 215588_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7234
p=0.4362
p=0.1264
p=0.7892
p=0.0409
p=0.0775
p=0.1004
p=0.1662
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

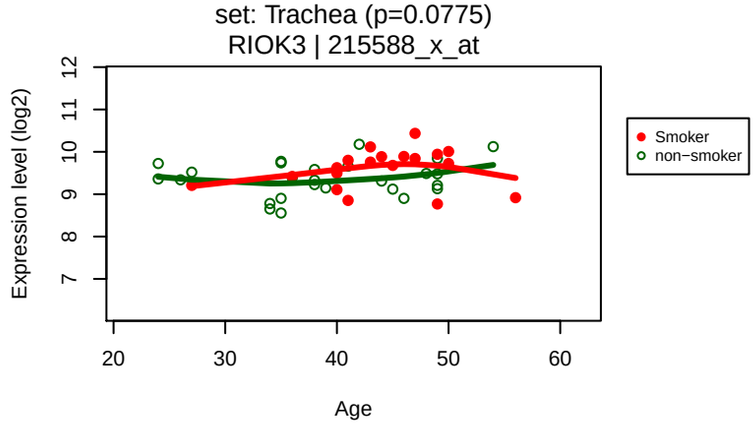

Gene expression ratio in Males
gene: RIOK3 | 215588_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5957
p=0.7748
p=0.1554
p=0.0658
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RIOK3 | 215588_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6713
p=0.0077
p=0.0524
p=0.8437
SmallAirway2
LargeAirway2
Trachea
Macrophage2

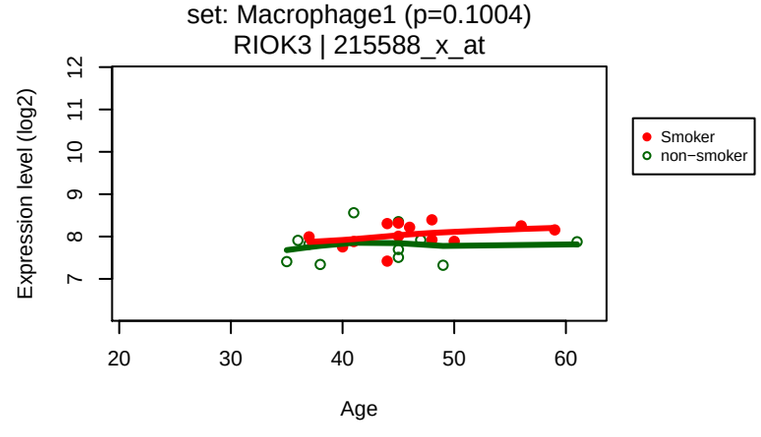

set: SmallAirway1 (p=0.7234)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker

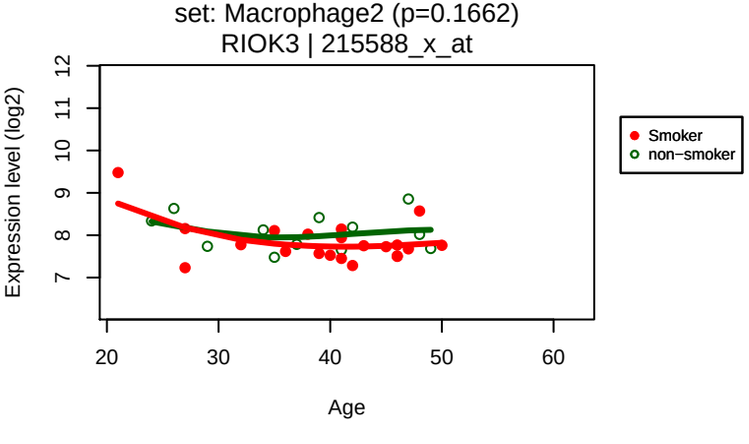

set: SmallAirway2 (p=0.4362)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker

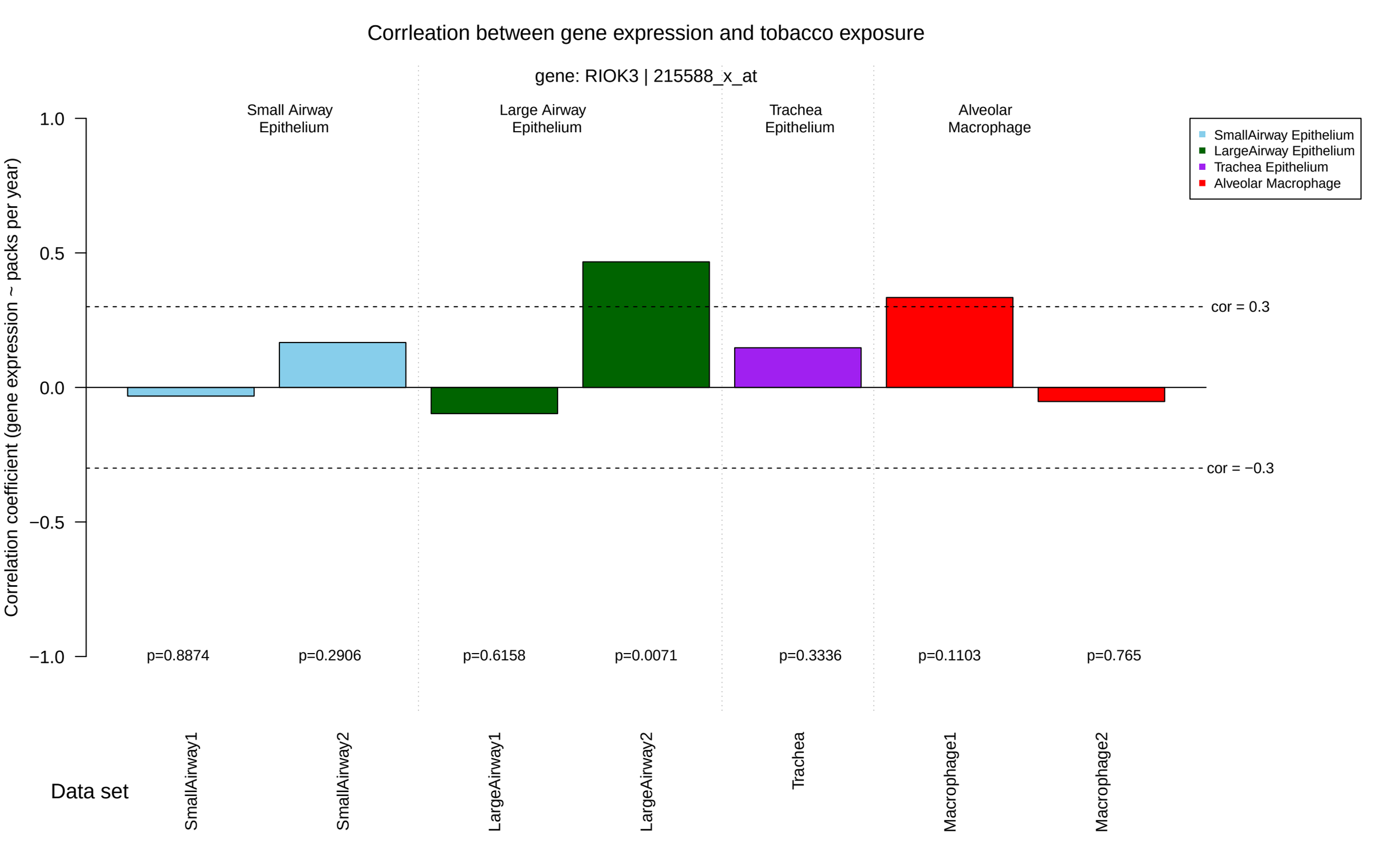

set: LargeAirway1 (p=0.7892)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0409)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0775)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1004)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1662)
RIOK3 | 215588_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RIOK3 | 215588_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8874
p=0.2906
p=0.6158
p=0.0071
p=0.3336
p=0.1103
p=0.765
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2